Arbitrary Direction Ship Detection in Remote-Sensing Images Based on Multitask Learning and Multiregion Feature Fusion

Qiangwei Liu, Xiuqiao Xiang, Zhou Yang, Yu Hu, and Yuming Hong

***Abstract*— Ship detection in remote sensing plays an important role in civil and military fields. Owing to the complex background and uncertain direction, ship detection is full of challenge by using the commonly used object-detection methods. In this article, a new framework for detecting the arbitrary direction ships is proposed based on the improvement in the Faster region-based convolutional network (R-CNN), in which the shape of the bounding box is described by three sides, namely, vertical side, horizontal side, and short side, respectively. The inclination of the ship is obtained by calculating the arc-tangent value of the vertical side to the horizontal side. First, the better performing ResNet-101 is adopted to extract features over an entire image, which are shared by the region proposal network (RPN) and the head network. Then, the multidirection proposal regions that may contain ships are generated by the RPN. Next, the global and local features of the proposal regions are combined as the whole features of the regions by a multiregion feature-fusion (MFF) module, which can provide more detailed information of the regions. Finally, the head network uses the whole features of the proposal regions for bounding-box recognition through multitask learning, including classification, regression, and incline direction prediction (left or right). The proposed method is tested and compared with other state-of-the-art ship-detection methods on two open remote-sensing data sets and some large-scale and real images. The experimental results validate that the proposed approach has achieved better performance.**

***Index Terms*— Convolutional neural network (CNN), multiregion feature fusion (MFF), remote-sensing image, rotation bounding box, ship detection.**

I. Introduction

As a traditional task of the coastal countries all over the world, ship detection and surveillance are widely used in ship search and rescue, fishing vessel management, territorial defense, oil-discharge control, illegal dumping, and sea-pollution monitoring [1]. With the rapid development of the remote-sensing technology, high-resolution remote-sensing images can present more detailed information of the ground objects, making it possible to identify and monitor ships effectively [2], [3]. Ship monitoring from the optical remote-sensing images is becoming a popular way due to the large amount of image acquisition and free opening.

A lot of meaningful research has been done on ship detection in the remote-sensing images. The process of these methods can be summarized as two steps: the extraction of the ship candidate regions and the discrimination of the ship targets, which is the general idea of object detection [4], [5]. In order to obtain the candidate regions, sliding window [6], saliency detection [1], [2], [7], and sea-land segmentation [3], [8]–[11] are usually adopted. After that, the recognition of the real ship is performed based on the ship template [6], [12], shape feature (such as aspect ratio, compactness, and convexness) [2], [3], [8]–[10], HOG feature [1], [7], Hu moments [11], and so on. These methods are severely disturbed by the background in the first stage, making it difficult for them to obtain all the ship candidate regions. Moreover, the features used in ship discrimination are designed by hand, which cannot achieve good robustness and generalization for ships with diverse shapes, scales, and angles [4]. Therefore, the accuracy and speed of ship detection in the remote-sensing images still need to be greatly improved.

At present, more and more attention has been paid to deep learning in the field of object detection, and a series of deep learning-based methods has been provided, such as region-based convolutional network (R-CNN) [13], Fast R-CNN [14], Faster R-CNN [15], YOLO [16], and single shot multibox detector (SSD) [17]. Compared with the traditional methods, deep learning can automatically learn the essential features from large amounts of image data and has higher accuracy and better robustness in object detection. However, in comparison with the vertical targets in the natural images, the directions of the ship targets in the remote sensing images are usually uncertain, since these images are taken from high altitude with an overlooking viewpoint. In addition, the remote-sensing images come from a larger field of view, leading to the background more complex, such as dense vessels, cluttered docks, clouds, and waves [2], [8]. The popular deep learning-based ship-detection methods [18]–[21] represent the position of the ship by the classical horizontal bounding box, which will surround more of the background, resulting in visual disturbances. Even worse, when the ship is close to each other, these methods will have serious missed detection due to the influence of nonmaximum suppression (NMS) [22], as shown in Fig. 1.

In order to solve the problems mentioned above, the detection method for the ship in any direction based on deep

Manuscript received September 4, 2019; revised January 19, 2020 and May 4, 2020; accepted May 27, 2020. Date of publication June 30, 2020; date of current version January 21, 2021. This work was supported in part by the State Scholarship Fund of China Scholarship Council under Grant 201806415025, and in part by the National Natural Science Foundation of China under Grant 61202051. *(Corresponding author: Xiuqiao Xiang.)*

The authors are with the School of Geography and Information Engineering, China University of Geosciences, Wuhan 430078, China (e-mail: xiangxq0005@126.com).

Color versions of one or more of the figures in this article are available online at https://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2020.3002850

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:19:10 UTC from IEEE Xplore. Restrictions apply.

learning is deeply studied in recent years. Zhang *et al.* [23] designed a ship-detection method named ship detection based on the convolutional neural network (S-CNN), in which region proposals are yielded through the ship model and the salient map, and then, the features of regions are automatically extracted by the convolutional neural network (CNN) for real ship identification. This method is inspired by the R-CNN, but it uses the improved region proposal scheme instead of the selective search algorithm [24], making it possible to obtain regions in any direction. Liu *et al.* [25] proposed the RR-CNN based on the Fast R-CNN, which uses the external algorithm ship rotated bounding box space method (SRBBS) [26] to get regional proposals and adds angle regression to determine the direction of the ship. These two methods have achieved good results, but both of them are non-end-to-end ship-detection methods and need to use other algorithms, such as saliency detection, image rotation, or segmentation, to provide region proposals; this directly affects the accuracy and speed of the overall method. Lin *et al.* [27] used the fully convolutional network (FCN) to detect the inshore ships in any direction and closely docked side by side. Through the combination of detection in deep path and accurate localization in shallow path, this method can effectively detect the foredeck/stern of the ship, but it is unable to get the direction prediction of the proposed candidates and needs to be further perfected. Zhang *et al.* [28] provided a region-based CNN method to detect ships. In this method, the small regions that may contain gathering ships or single ship are obtained and cropped through water-land classification and morphological analysis, and then, each region is rotated in the main direction to put the ship in a vertical state according to the HOG feature. Finally, all the rotated regions are detected one by one by the improved Faster R-CNN. This method is very effective for dense ships, but it has to carry out cropping, rotation, and detection for each small region; this is obviously not as simple and fast as directly predicting the inclined bounding box of the ship in a larger range. Li *et al.* and Yang *et al.* [29], [30] used the modified Faster R-CNN to detect the inclined ships by predicting not only the center coordinates (x, y) and the length width (H, W) of the bounding boxes but also the angle θ that directly represents the inclination and direction information of the ships. This method realizes end-to-end ship detection and achieves better results than the detection methods based on the horizontal bounding box. However, the learning of the angle θ is affected by the distribution of the angle ground truth in the data set, and for some special angles, such as 90° and −90°, the features are so similar that it is difficult for the model to learn stable features [31], [32].

In this article, we propose a novel framework to detect the ship in any direction based on the Faster R-CNN architecture. The Faster R-CNN describes the shape of the horizontal rectangular bounding boxes of the target by the length of the vertical and horizontal sides. Inspired by this, we add the prediction of the third side (called short side) to produce the inclined bounding boxes. The main contributions of this article are listed as follows.

(a) (b)

Fig. 1. (a) Confused visual effect because of the dense overlap between the bounding boxes. (b) Missed detection due to excessive overlap between the bounding boxes.

1) The inclined bounding box of the ship is determined by the regression of the three sides, which transforms angle regression into side regression and avoids the problems caused by the direct prediction of the angle.
2) The bounding box of the ship target is acquired through end-to-end multitask learning, including classification, the regression of the bounding box, and the prediction of the incline direction (left or right).
3) A multiregion feature-fusion (MFF) module is constructed to provide more target details, and it is helpful for the accurate identification of the ship and bounding box regression.

The rest of this article is organized as follows. Section II introduces the framework and details of the proposed method. Section III presents the experiments conducted on some open remote-sensing image data sets to verify the effectiveness of the proposed approach. Finally, Section IV discusses and concludes the work of this article.

II. Proposed Method

Until now, the Faster R-CNN is still one of the optimum models in the field of object detection. In this article, the classical Faster R-CNN is improved and applied to arbitrary-oriented ship detection in the remote-sensing images. The framework of the proposed method consists of a backbone network, region proposal network (RPN), MFF module, and head network, as described in Fig. 2.

We take the better performing ResNet-101 [33] as the backbone network of the framework. The backbone network is used to extract features over an entire image, which will be shared by the RPN and the head network. The region proposals are generated automatically by the RPN, which are no longer just in the horizontal direction but multiple directions. After that, the global region and the local regions are obtained by calculating the minimum bounding rectangle of each region proposal, and the features of these regions are combined as the overall features of each region proposal and transmitted to the head network for more accurate classification and location. The output of the head network has three tasks: category classification, bounding box regression, and incline direction prediction. The loss for each task is calculated, and their sum is used as the total loss for back propagation to train the network model. When ship detection is carried out, the center coordinates (x, y), the length of three sides (W_h, W_w, and H),

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:19:10 UTC from IEEE Xplore. Restrictions apply.

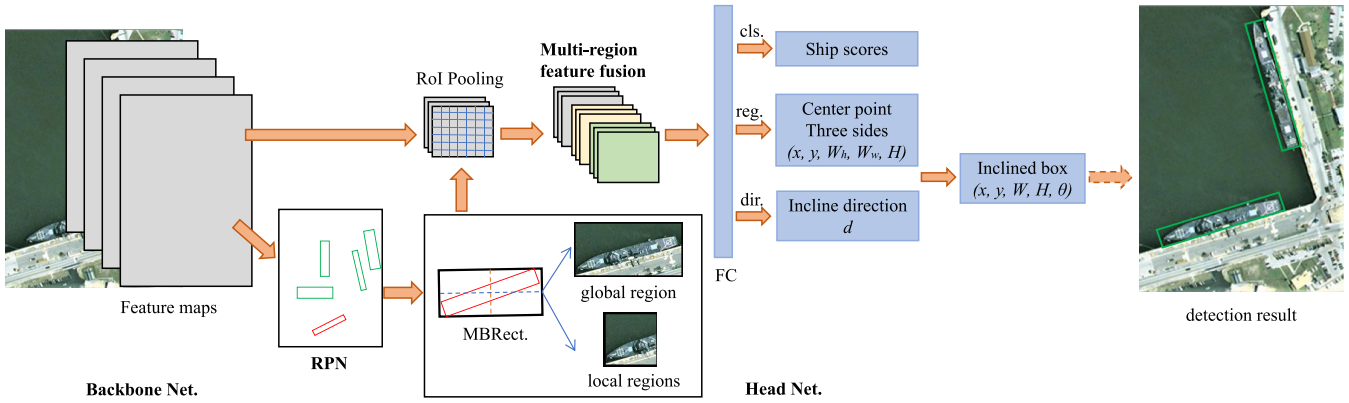


Fig. 2. Overall framework of our proposed ship-detection method.

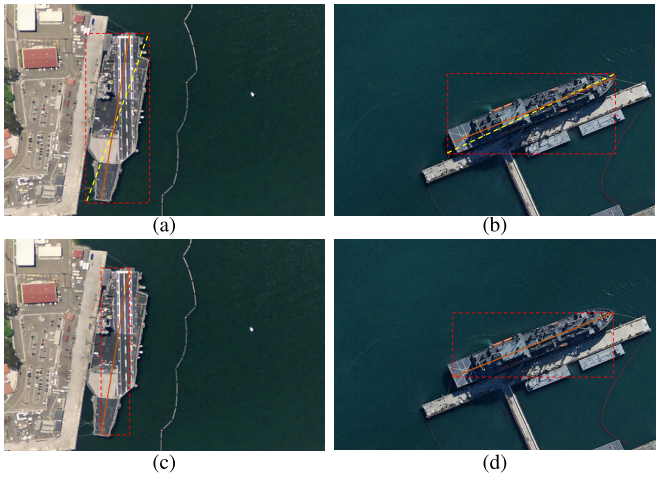


Fig. 3. Orange solid line: center line of a ship's hull. Yellow dotted line: diagonal of the horizontal bounding box. (a) and (b) Difference in the length and deviation in the direction between the ship and diagonal of the horizontal bounding box. (c) and (d) Rectangle we use, where the hull is coincident with the diagonal of the rectangle.

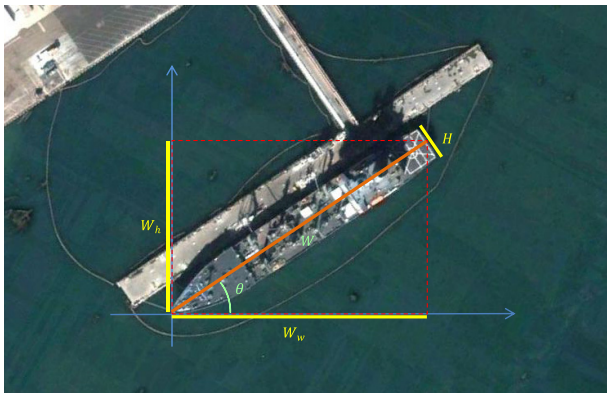


Fig. 4. Shape of the inclined bounding box is described by three sides: vertical side W_h, horizontal side W_w, and short side H.

and the incline direction (left or right) are predicted through the network, so that the specific position of the ship is detected.

A. Design of Three Sides

As we know, the first requirement for object detection is the accurate position usually marked by the horizontal rectangular bounding box, as the Faster R-CNN does. If this bounding box is used for the detection of a ship with a slender shape, the target is approximately located on the diagonal of the bounding box, and its length is roughly the same as that of the diagonal. However, in some cases, the hull is nearly vertical (or horizontal) or the ship has a relatively wide bow or stern, and the hull will deviate from the diagonal or their length will no longer be the same, as shown in Fig. 3(a) and (b). Based on this, we adopt a new rectangle, which no longer surrounds the whole ship, but the center line of the hull. This ensures that the length and angle of the rectangle's diagonal are the length and angle of the ship. Fig. 3(c) and (d) shows two examples of our improved rectangles.

In above case, the ship's shape will be determined if the width of the ship is also predicted. In other words, the inclined bounding box of the ship can be described by predicting the length of the three sides, as illustrated in Fig. 4, and the three sides are all indicated by the yellow line. The short side expressed by H corresponds to the width of the ship, that is, the height of the rectangle in geometry. W_h and W_w represent the vertical and horizontal sides of the improved rectangle, respectively. Thus, the length W and angle θ of the ship can be calculated as follows:

$$W = \left(W_h^2 + W_w^2\right)^{1/2} \tag{1}$$

$$\theta = \arctan(W_h/W_w), \quad \theta \in (0°, 90°). \tag{2}$$

As mentioned above, the inclination of the ship is calculated by the arc-tangent operation of W_h and W_w. Because both W_h and W_w are positive, the obtained angle value can only be distributed between 0° and 90°, while its real direction is unclear, that is, one angle value corresponds to two possible cases: left or right direction, as illustrated in Fig. 5. Therefore, it is necessary to judge the incline direction (left or right) of the ship to determine the real case.

To tackle this issue, we add a branch to the head network to predict the incline direction of the ship, which is divided into two types, left (0°, 90°) and right (−90°, 0°), and recorded as 0 and 1, respectively. Thus, the inclined ship can be clearly expressed as a tuple with structure (x, y, W_h, W_w, H, d), where (x, y) indicates the geometric center of the bounding box, and W_h and W_w represent the vertical and horizontal sides, respectively. H always denotes the ship's short side

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:19:10 UTC from IEEE Xplore. Restrictions apply.

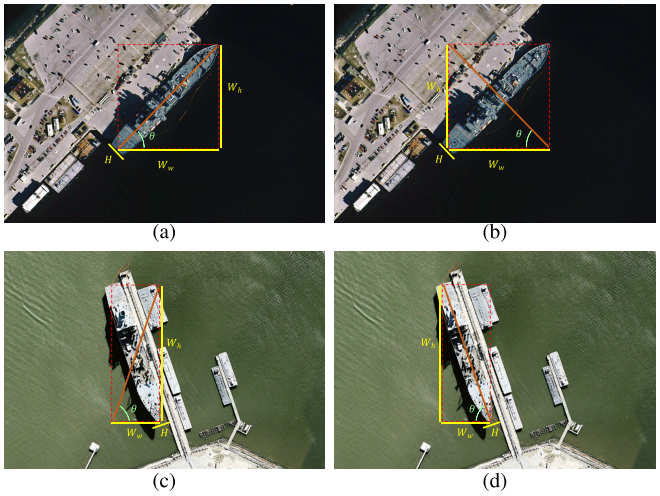


Fig. 5. One tuple (W_h, W_w) corresponds to two incline cases, which are inclined at the same angle,but in the opposite direction (left or right). In the example of the first line, it is clear that result (a) is correct and result (b) is wrong, while for the second line, result (c) is wrong and result (d) is correct.

and d is the incline direction of the ship with a value 0 or 1. Combined with formula (2), the true angle value can be obtained as follows:

$$\theta' = (-2d + 1)\theta. \tag{3}$$

Since the hull is usually long, the incline direction is easy to identify, which is a simple binary classification problem. The addition of this task will not significantly increase the computing time of the model, because the judgment of the incline direction shares the same features with the classification and bounding box regression. Note that the network used in our work is improved on the Faster R-CNN, which is named the three-side prediction network (TPN) in this article.

B. Multidirection Anchors

In the traditional RPN [15], the anchors are the horizontal rectangles with a small aspect ratio. While for slender ships, the horizontal anchor is not suitable, because it will contain a large area of unnecessary background, particularly when the ship is inclined and dense. In order to better detect a ship with arbitrary direction, the common rotating anchor approach is adopted and the anchors in 12 directions are designed in this article: left (0°, 15°, 30°, 45°, 60°, 75°) and right (−90°, −75°, −60°, −45°, −30°, −15°). According to the specific characteristics of the ship's shape, the aspect ratio is designed in two forms 1:4 and 1:7. For the images with submeter resolution, four area scales (64^2, 128^2, 256^2, 384^2) are employed. That is, there are four scales of anchors at each angle, and the anchors of each scale have two different aspect ratios, so that $12 \times 4 \times 2$ anchors will be generated at each feature point. The determination of these parameters will be analyzed detailedly in Section III, and the partial anchors are shown in Fig. 6.

It is worth noting that the anchors inclined to the left (red) are determined by (x, y, W_h, W_w, H, 0) and the anchors inclined to the right (blue) are controlled by (x, y, W_h, W_w, H, 1). The number of multidirection anchors at one point increases by five times compared with simple horizontal anchors, and these anchors overlap with each other, so it is necessary to eliminate the anchors beyond the boundary and those with high overlap rates and low scores. For this purpose, nonmaximum suppression (NMS) is usually used. First, the Intersection-over-Union (IoU) between the horizontal bounding boxes is calculated, then the bounding box with a low score and IoU greater than the threshold is removed, and only one optimal or a few bounding boxes with a low overlap are retained finally. Obviously, for two or more targets close to each other, traditional NMS may remove the candidate boxes of other adjacent targets, resulting in missed detection. To decrease the overlap ratio of the adjacent candidate regions, Skew IoU [34] is adopted to reduce their influence on each other in our model, which is calculated as follows:

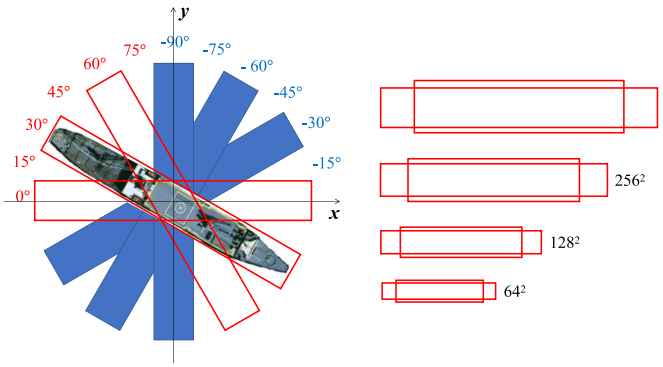


Fig. 6. Angle design of the inclined anchors in the proposed method. For the sake of clarity, some anchors are not displayed directly, but the angle value is marked. There are four scales of anchor at each angle, and the anchor of each scale has two aspect ratios.

$$\begin{aligned} \text{IoU} &= \frac{\text{Area}_{\cap}}{\text{Area}_{\cup}} \\ &= \frac{S_{\text{incB}_1} \cap S_{\text{incB}_2}}{S_{\text{incB}_1} + S_{\text{incB}_2} - S_{\text{incB}_1} \cap S_{\text{incB}_2}} \end{aligned} \tag{4}$$

where S_{incB_i} represents the area of the inclined bounding box incB_i ($i = 1, 2$), likewise for $S_{\text{incB}_1} \cap S_{\text{incB}_2}$. $\cap$ and $\cup$ denote the intersection operation and the union operation, respectively.

C. Multiregion Feature Fusion

Multisource remote-sensing image fusion can make the image quality better and is often used to improve the image spatial resolution [35]. Similarly, the fusion of the multiregion feature information of the target can enhance the robustness of the feature for accurate classification and localization, since they can provide more local details of the target [36], [37]. In this article, we adopt this processing strategy to construct a multiregion feature-fusion (MFF) module and combine it with TPN (i.e., MFF-TPN) as the overall architecture of ship detection. This is a modular network that can also be inserted into other object-detection frameworks.

We first obtain the candidate regions in any direction through the RPN and calculate the horizontal minimum bounding rectangle (MBR) for each region, which is a global region

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:19:10 UTC from IEEE Xplore. Restrictions apply.

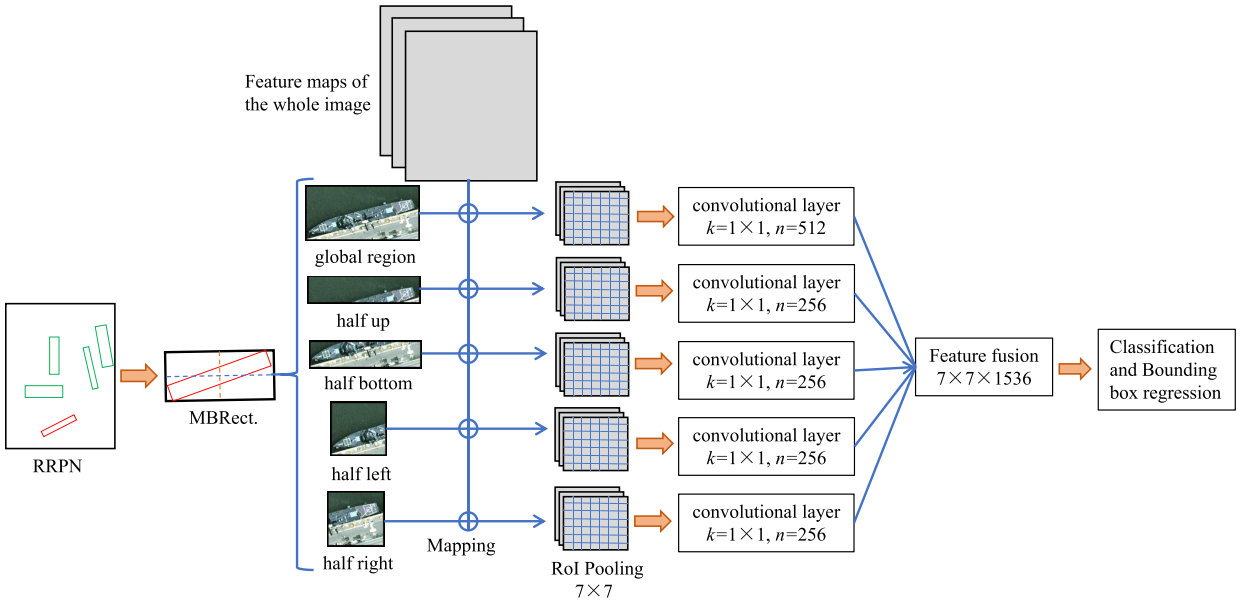


Fig. 7. Details of the multiregion feature-fusion module.

that covers the whole of the potential target. Then, we divide the rectangle into four local regions: the upper/lower/left/right half part, each containing only the head or tail of the ship. Note that the features of these regions are not cropped directly from the original image, but like the Fast R-CNN, the features of the corresponding region are gained from the feature map that is generated by the convolution of the whole image. After that, these features are unified to the same size by RoI pooling and convoluted again to obtain the unique features of each region. We use 1×1 kernel to realize convolution, which can integrate the features of different channels and increase the nonlinearity of the features without losing the resolution [38]. Finally, these features are connected together in the channel direction as the whole features of the region for classification and bounding box regression. Fig. 7 shows the details of the multiregion feature-fusion module. This scheme considers the global and local information of the candidate regions, which makes the feature expression richer and more robust and, thus, more sensitive to the category and location of the target [37].

D. Multitask Loss

In the region proposal phase, the RPN loss is calculated in the same way as the Faster R-CNN. In the head network phase, because we have added a new task to determine the incline direction, the loss in this stage is the sum of the loss of the three tasks: classification, bounding box regression, and incline direction prediction, defined as

$$L(p, l, v^*, v, q, d) = L_{\text{cls}}(p, l) + \lambda_1 l L_{\text{reg}}(v^*, v) + \lambda_2 l L_{\text{dire}}(q, d) \quad (5)$$

where λ_1 and λ_2 are the balancing parameters that control the tradeoff between the three terms, and all experiments use $\lambda_1 = \lambda_2 = 1$ in this article. l is the indicator of the class label, ship is labeled as 1, and background is labeled as 0, which means that the bounding box regression and the prediction of the incline direction are activated only for positive proposals, and p is the probability over the ship and background classes computed by the softmax function, so $L_{\text{cls}}(p, l) = -\log p_l$ is calculated as the classification loss for true class l. d represents the incline direction of the ship ($d = 1$ for right and $d = 0$ for left), and this is also a binary classification problem. Probability q of inclining to the left or to the right is still computed by the softmax function and $L_{\text{dire}}(q, d) = -\log q_d$.

If $v = (v_x, v_y, v_{W_h}, v_{W_w}, v_H)$ is the predicted tuple for the ship label, $v^* = (v_x^*, v_y^*, v_{W_h}^*, v_{W_w}^*, v_H^*)$ denotes the ground truth, and then the regression loss is calculated by a robust loss function (smooth L_1) [14], which is defined as

$$L(v^*, v) = \sum_{i \in \{x, y, W_h, W_w, H\}} \text{smooth}_{L_1}(v^* - v) \quad (6)$$

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise.} \end{cases} \quad (7)$$

In the Faster R-CNN, the center coordinates and two sides are used to determine the horizontal bounding box of the target, while the proposed method describes the ship's inclined bounding box by the center coordinates and three sides. They are basically the same in the principle of the bounding box regression, so we may take a similar idea to calculate the scale-invariant parameterization tuple v, v^* as follows:

$$v_x = \frac{x - x_a}{W_a}, \quad v_y = \frac{y - y_a}{H_a}$$
$$v_{W_h} = \log \frac{W_h}{W_{h_a}}, \quad v_{W_w} = \log \frac{W_w}{W_{w_a}}, \quad v_H = \log \frac{H}{H_a} \quad (8)$$
$$v_x^* = \frac{x^* - x_a}{W_a}, \quad v_y^* = \frac{y^* - y_a}{H_a}$$
$$v_{W_h}^* = \log \frac{W_h^*}{W_{h_a}}, \quad v_{W_w}^* = \log \frac{W_w^*}{W_{w_a}}, \quad v_H^* = \log \frac{H^*}{H_a} \quad (9)$$

where $W_a = (W_{h_a}^2 + W_{w_a}^2)^{1/2}$, variables x, x_a, and x^* are for the predicted box, anchor box, and ground-truth box, respectively (likewise for y, W_h, W_w, W, H).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:19:10 UTC from IEEE Xplore. Restrictions apply.

III. Experiment

In this section, the proposed method is evaluated experimentally and the detection performance is compared with that of the classical Faster R-CNN [15], R-DFPN [30], R2CNN [31], and RRPN [34]. All experiments are performed on an Intel Core i5-7300HQ CPU machine with 8 GB of RAM, and all detection methods are carried out using the Python language on the deep learning framework, TensorFlow.

A. Remote-Sensing Data Set

1) HRSC2016 Data Set: We evaluate the performance of our method on High Resolution Ship Collections 2016 (HRSC2016)[1] [39], which is a public remote-sensing data set specially released for ship detection. It contains 1070 images with 2976 ship targets collected from Google Earth and 2886 ships are in harbors with the complex background, which makes the ship detection become more difficult. These ships include war craft, aircraft carrier, cargo ship, and so on, collected with large variations of rotation, scale, shapes, and appearance. For each ship, two kinds of ground truths are provided: horizontal bounding box and inclined bounding box. In the experiment, we exclude hovercraft and submarine samples and retain 996 images. The image resolutions are between 0.4 and 2 m, the image sizes range from 300×300 to 1500×900, and most of the image sizes are larger than 1000×600.

2) NWPU-Ship Data Set: For further verifying the validity of our method, we carry out experiments on another remote-sensing data set named Northwestern Polytechnical University very high-spatial resolution-10 (NWPU VHR-10)[2] [40]. This is a challenging geospatial object-detection data set with ten categories, which can be used for single class and multiclass detection. These images are taken from Google Earth with spatial resolutions ranging from 0.5 to 2 m. We only keep the images containing the ship, a total of 54 images. Because the ground truth of the data set is only the horizontal bounding box, we relabel these images and get 314 ship samples, called NWPU-ship data set. Compared with the HRSC2016 data set, the view of the images is larger and the ship size is smaller, most of which are scattered ships on the sea.

3) Real and Large-Scale Remote-Sensing Images: We also collect four large-scale optical remote-sensing images with a minimum size of 3000×2000 and a maximum size of 8310×5083. There are two images from GeoEye-1 and DEIMOS-2, with resolutions of 0.5 and 0.75 m, respectively. The other two are Google images with a resolution of 0.6 m.

In order to increase the amount of training data and improve the diversity of samples, the original training data are cropped by setting the overlap rate to 85%, and the long side of the cropped images is controlled not more than 1000. Then, the cropped images are further expanded by rotating, blurring, and adjusting contrast. Finally, the HRSC2016 data set has a total of 4276 training images and 2936 test images and the NWPU-ship data set has a total of 2580 training images and 896 test images. All the detection methods are trained on the HRSC2016 data set to obtain stable models, and then the test is carried out. For the NWPU-ship data set, the trained models are fine-tuned on the training images before testing. For the detection on large-scale images, the HRSC2016 trained model is adopted directly.

[1]http://www.escience.cn/people/liuzikun/DataSet.html

[2]http://jiong.tea.ac.cn/people/JunweiHan/NWPUVHR10dataset.html

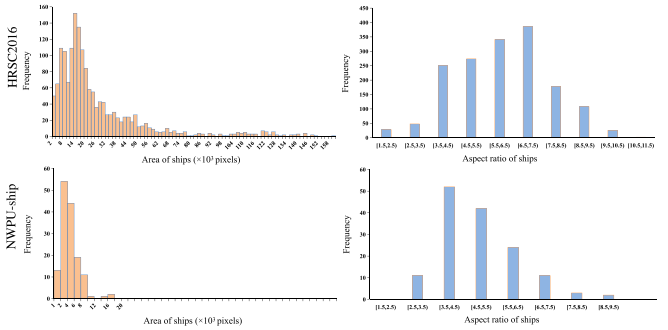


Fig. 8. Frequency distribution of the ships' area and aspect ratio in the training images of two data sets.

B. Training Details

Faster R-CNN, R2CNN, RRPN, and our method are all initialized by the pretrained ResNet-101 model, while the R-DFPN is a fine-training of the original model. On the HRSC2016 data set, each method is trained 1.0×10^5 iterations and the initial learning rate is set to 0.001. After 4.0×10^4 and 8.0×10^4 iterations, the learning rate decreased to 0.1 times of the original. The momentum optimizer is chosen, and the weight decay and momentum are 0.0001 and 0.9, respectively. In addition, to select a suitable anchor for the ship detection, the area and aspect ratio of the ships in the training images are analyzed carefully, and the results are shown in Fig. 8. Fig. 8 demonstrates that for the ship targets with resolution between 2 and 0.4 m, most of the ships' area is concentrated between 2×10^3 and 70×10^3 pixels, and there are also a few ships with larger area. The aspect ratio of most ships ranges from 3.5 to 7.5, which is the size of a conventional ship [41]. According to the analysis results, we have designed multidirection anchors with four scales (64^2, 128^2, 256^2, and 384^2) and two aspect ratios (4.0 and 7.0).

For RPN training, the training samples are determined as follows [15]: when IoU between an anchor and the ground truth is greater than 0.7, it will be treated as a positive sample. While the anchors with an IoU less than 0.3 are defined as negative samples, other anchors will not participate in the training. The total number of positive and negative samples is 256 per mini-batch, and the ratio of positive to negative samples is set to 0.5.

For all the proposal regions generated by the RPN, the threshold of the NMS is set to 0.7 so that regions with higher overlaps can be removed, and only 1000 better quality proposal regions are selected for subsequent networks (during the test phase, this value is reduced to 300 for faster detection). In the final display of the detection results, only one optimal bounding box is retained in one position by setting a lower

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:19:10 UTC from IEEE Xplore. Restrictions apply.

Fig. 9. (a) Detection result located by horizontal bounding box. (b) Detection result located by inclined bounding box. Compared with the horizontal bounding box, IoU is more sensitive to the change in the inclined bounding box. Green box: ground truth. Red box: predicted bounding box.

NMS threshold 0.2. In addition, we still use the traditional RoIPooling with 7×7 pooled size to unify the size of the proposals as well as keep more useful semantic and spatial information.

C. Testing Performance

Based on the test images of the two data sets, the ship detection is performed by the proposed method and compared with other excellent approaches. To evaluate quantitatively the detection performance of these detection methods, we take the general indicators for further analysis. The evaluating indicators mainly include Recall (R), Precision (P), Average precision (AP), F1-Score (F1), and detection time. The detection time is evaluated by measuring the average time of detecting an image. Recall is the ratio of the true target detected to the total target, and Precision is the proportion of the true target detected to all the detection results. Average precision is defined as the mean precision on a series of recall levels [42]. F1-Score is the harmonic mean of Precision and Recall [43], which is defined as

$$F1 = 2PR/(P + R). \tag{10}$$

Using the test images, the F1-Score under the specific IoU threshold can be calculated. In general, the detection bounding box with an IoU overlap ≥ 0.5 is considered as a positive [13], [14]. We hope to raise the IoU threshold to evaluate methods by stricter criteria. However, the small and slender shape of the ship makes IoU more sensitive to the change in the inclined bounding box, even a small deviation will lead to a great reduction in IoU, resulting in unreasonable judgment [30], as shown in Fig. 9. Compared with Fig. 9(a), the detection result of Fig. 9(b) is less different from the ground truth, but the calculated IoU of the inclined bounding box is smaller. When the IoU threshold is 0.7, the detection result of Fig. 9(b) will be considered as error detection, resulting in a reduction in precision and recall; this is clearly unreasonable. Considering that the high IoU seriously increases the missed detection of the inclined bounding box, while the low IoU is not strict in the position constraint, we set the IoU threshold to 0.6 to evaluate the performance of each method.

1) Analysis of the Incline Direction Prediction: As one of the output tasks of detection network, the judgment of incline direction has an important influence on the determination of the ship targets' bounding box. The incline direction prediction is a binary classification task, and only when the real ship target is detected does the incline direction make sense. In our experiment, we compared the annotated incline direction and predicted the incline direction of all recalled ship targets and got an accuracy of 99.3%, which demonstrates that the incline direction of the ship target is easy to judge correctly, since the ship target has a relatively large aspect ratio and the long spindle owns obvious characteristic.

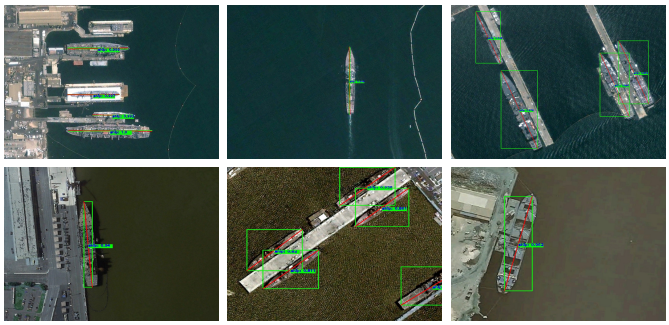


Fig. 10. Visualization of the long side and inclination angle obtained by the proposed method. The green bounding box represents the vertical and horizontal sides of the predicted three sides, and the red line is the long side calculated from them. It can be seen that the red line fits the length and direction of the ship's spindle very well.

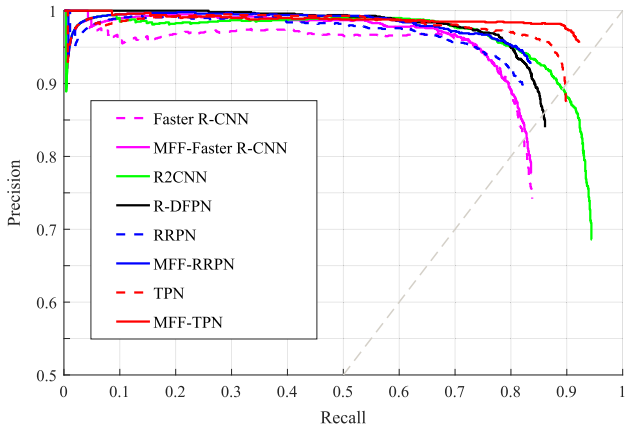


Fig. 11. Precision–recall curve of different methods on the HRSC2016 data set. The intersection of gray dotted lines with other curves is the equilibrium point of each method. The higher the position of the equilibrium point, the better the detection performance.

2) Analysis of the Vertical- and Horizontal-Side Prediction: The proposed method determines the bounding box of the ship in any direction by predicting three sides, including vertical side, horizontal side, and short side. Since the vertical and horizontal sides are used to calculate the inclination angle and long side of the ship by the arc-tangent operation and Pythagorean theorem, the prediction accuracy of these two sides is very important. Some detection results of these two sides are visualized in Fig. 10. It can be seen that no matter how the ship inclines, the angle and long side calculated by the predicted vertical and horizontal sides are basically consistent with the direction and length of the ship's spindle, which shows good performance of the proposed method and illustrates the feasibility of solving the ship's inclination angle by predicting the vertical and horizontal sides.

3) Evaluation of MFF: When constructing the multiregion feature-fusion (MFF) module, the reasonable number of convolution kernels for each region needs to be considered.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:19:10 UTC from IEEE Xplore. Restrictions apply.

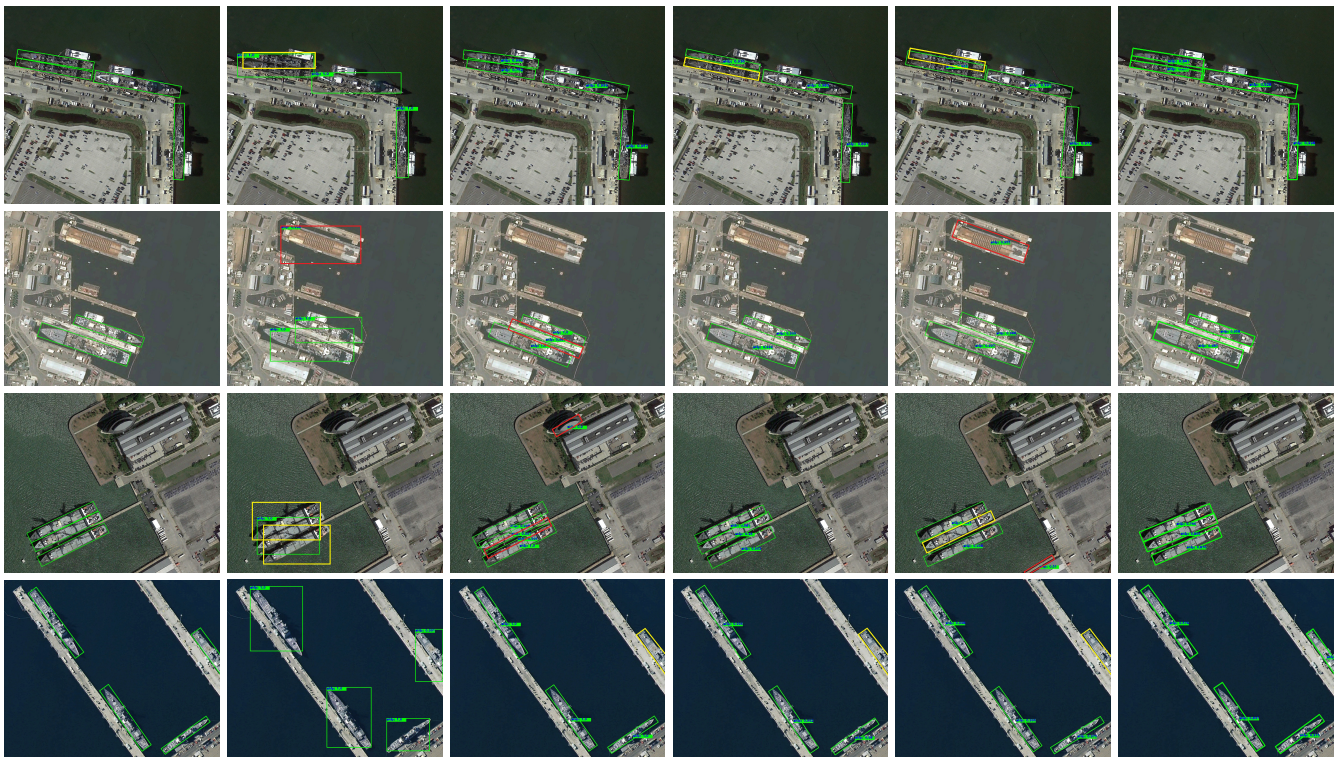


Fig. 12. Some detection results of various methods on the HRSC2016 data set. In the images, the green bounding box represents the correct result, the yellow box indicates the missed detection, and the red box denotes the false detection. (a) Ground truth. (b) Faster R-CNN. (c) R2CNN. (d) RRPN. (e) R-DFPN. (f) Ours.

TABLE I
PERFORMANCE COMPARISON OF MFF WITH DIFFERENT NUMBERS OF KERNELS

kernel number	P	R	F1	Time(s)
(256, 256, 256, 256, 256)	0.962	0.894	0.927	0.367
(512, 512, 512, 512, 512)	0.958	0.914	0.935	0.371
(512, 256, 256, 256, 256)	0.968	0.912	0.939	0.367

TABLE II
PERFORMANCE COMPARISON OF THE METHOD WITH AND WITHOUT MFF

Method	P	R	F1	mAP	Time(s)
Faster R-CNN	0.910	0.790	0.846	0.790	0.331
MFF-Faster R-CNN	0.905	0.801	0.850	0.800	0.333
RRPN	0.902	0.820	0.859	0.803	0.340
MFF-RRPN	0.937	0.829	0.880	0.825	0.345
TPN	0.951	0.876	0.912	0.887	0.362
MFF-TPN	0.968	0.912	0.939	0.912	0.367

We compared several different sets of parameters and determined the specific parameters of MFF used in this article according to the results shown in Table I. In order to verify the effectiveness of MFF, we also insert the designed MFF module to the architecture of the Faster R-CNN and RRPN. Table II lists the corresponding precision, recall rate, highest F1-score, and detection time for various approaches with and without the MFF module. It can be clearly seen from Table II that the precision and recall of the method with the MFF module outperform the original method by 1%–3%, which fully confirms that the MFF module is helpful to improve the performance of the detection method. In addition, the MFF module is almost no time-consuming, since this process directly uses the feature map generated by the previous network.

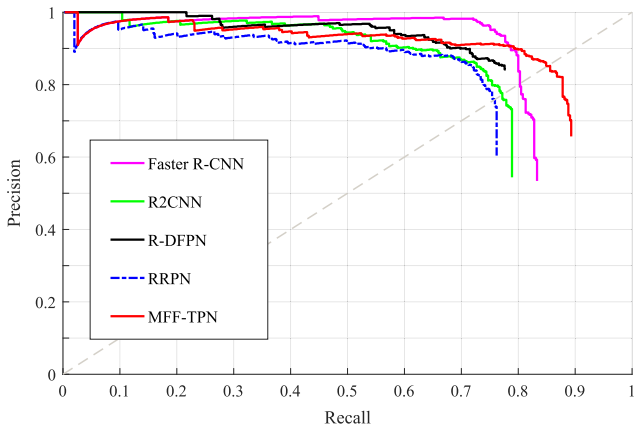


Fig. 13. Precision–recall curve of different methods on the NWPU-ship data set.

4) Comparisons With Other Detection Methods:

a) On HRSC2016 Data Set: In order to compare the performance of each method, the precision–recall curve of each method is drawn intuitively, as illustrated in Fig. 11. Table III quantitatively shows the detection performance of each method on HRSC2016. The experimental results show that the proposed method MFF-TPN is much better than other methods and achieves the most competitive results: 96.8% for Precision, 91.2% for Recall, and 93.9% for F1-Score. Compared with the least time-consuming Faster R-CNN, the detection time of the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:19:10 UTC from IEEE Xplore. Restrictions apply.

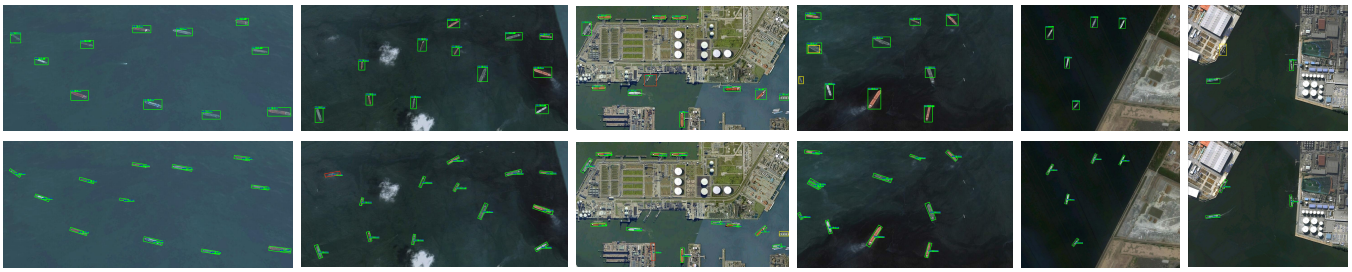


Fig. 14. Some detection results of (Top) Faster R-CNN and (Bottom) proposed method on the NWPU-ship data set. In the images, the green bounding box represents the correct result, the yellow box indicates the missed detection, and the red box denotes the false detection.

TABLE III
PERFORMANCE COMPARISON OF EACH DETECTION METHOD ON HRSC2016

Method	P	R	F1	Time(s)
Faster R-CNN	0.910	0.790	0.846	**0.331**
R2CNN	0.918	0.871	0.894	0.633
R-DFPN	0.921	0.833	0.875	0.460
RRPN	0.902	0.820	0.859	0.340
MFF-TPN	**0.968**	**0.912**	**0.939**	0.367

TABLE IV
PERFORMANCE COMPARISON OF EACH DETECTION METHOD ON NWPU-SHIP

Method	P	R	F1	Time(s)
Faster R-CNN	**0.928**	0.777	0.846	0.427
R2CNN	0.859	0.727	0.788	0.422
R-DFPN	0.853	0.776	0.813	**0.365**
RRPN	0.853	0.720	0.781	0.464
MFF-TPN	0.861	**0.849**	**0.855**	0.443

proposed method is also not significantly increased due to that a large number of low-quality region proposals are filtered out and multiple tasks share features.

Fig. 12 shows some results of these detection methods when the score threshold is 0.5. The experimental results validate the advantages of the inclined region proposals in the detection of dense ship with any direction. Because of the horizontal anchors used by the Faster R-CNN in the region proposal phase, some proposals containing the target are filtered out by NMS, which reduces the recall of the Faster R-CNN. The detection performance of the R2CNN that outputs both horizontal and inclined bounding boxes is better than that of the Faster R-CNN that only outputs the horizontal bounding boxes. Multiple tasks share the extracted features, which makes the features have a strong ability of abstraction, and multitask training can let the network learn some features that are not easy to obtain under a single task. This enables the model to achieve a better generalization and higher accuracy [44]. However, the R2CNN is also negatively affected by NMS and inferior to our method, since the R2CNN still does not change the direction of the anchor in the first phase. Furthermore, as shown in Fig. 12, it is easy for the R2CNN to consider the narrow area as a ship, which greatly reduces the detection accuracy. The R-DFPN should be good to improve the detection precision, since it combines the low-level location information and high-level semantic features. Nevertheless, the R-DFPN is susceptible to background (such as roofs and container piles) and easy to cause false alarm. In addition, the R-DFPN often misses the detection of ships close to each other, and this reduces the recall rate of the method to a certain extent, which is proven in our experiments and basically consistent with the image-detection results provided in the original article [30]. Our method reduces the impact between the dense ships due to the use of inclined bounding boxes and is more robust to complex background and incomplete ships (such as only the bow or stern in the image) by using more local and global region features, as illustrated in Fig. 12.

b) On NWPU-Ship Data Set: When fine-tuning the NWPU-ship data set, we remove the large-scale 384^2 while adding a smaller scale 32^2 for each method, and keep the other parameters unchanged. Then, the precision and recall rate of each method are evaluated in detail, as shown in Table IV, and the corresponding precision–recall curve is shown in Fig. 13.

The proposed approach has the highest F1-Score and recall rate, but its precision is lower than that of the classical Faster R-CNN. Through the analysis of the test results, we think that there are two reasons. First, the ships in the NWPU-ship data set are basically scattered on the sea surface and there is no excessive overlap between the bounding boxes of ships, so the advantages of the inclined bounding box cannot fully be reflected. Second, the ship targets in this data set are smaller, while the IoU of the small ship is more sensitive to the change in the inclined bounding box compared with the horizontal bounding box, as demonstrated in Fig. 9. In addition, some detection results of the proposed method and Faster R-CNN are shown in Fig. 14, from which it can be seen that although both methods have detected almost all the ships, the inclined bounding box is clearer in terms of position accuracy.

c) On Real and Large-Scale Remote-Sensing Images: Most of the ships in these images are docked at harbors, and the complex land background and the close proximity of the ships make it great challenge to detect the ship. Fig. 15 displays the detection results of our method using the model trained on the HRSC2016 data set, which validates the effectiveness of the proposed method in the real and large-scale images. Furthermore, the number of false detection, missed detection, and accurate detection of each method on these images is analyzed and the statistical result is shown in Fig. 16. Compared with other methods, the proposed method has relatively lower false detection and missed detection.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:19:10 UTC from IEEE Xplore. Restrictions apply.

Fig. 15. Detection results of the proposed method on large-scale images. (a) Keelung, China. (b) Yokosuka, Japan. (c) Vancouver, Canada. (d) Gwangyang, South Korea. Details can be viewed by magnification.

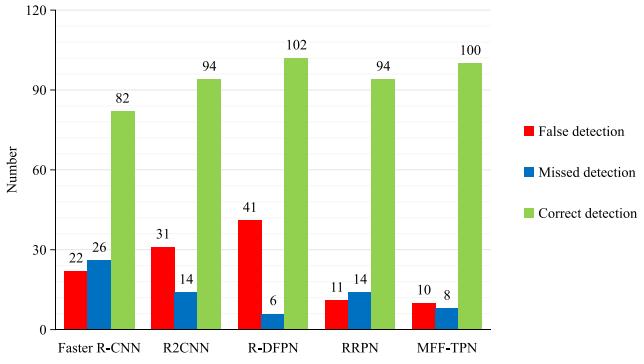


Fig. 16. Detection results of all methods on four lager scale images.

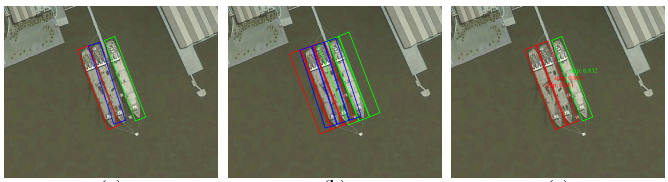


Fig. 17. (a) Detection results of MFF-TPN. (b) New region proposals generated through translation. (c) Final result obtained by reclassification.

D. Subsequent Processing

The ship parked at the dock usually presents a parallel distribution; this brings a challenge to the prediction of the inclined bounding box. The dense arrangement of the ships often results in inaccurate prediction of the angle or edge length of a single ship, as displayed in Fig. 17(a). In general, the ships docked side by side have the same size and inclination angle. Based on this, both sides of the bounding box outputted by the head network are translated n times with their own width as the step size, so as to obtain new candidate regions; this is called The Second Region Proposal. In the experiment, we set the translation number $n = 1$, that is, the translation to both sides of the bounding box is performed once, as presented in Fig. 17(b). After that, we do not regress the specific parameters of the bounding box but only carry on the object recognition to the new region proposals by reusing the weights of the head network, since the correct bounding boxes have been well predicted in the head network stage, and the training process need not be performed again. Finally, the more suitable bounding boxes with higher scores are retained

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:19:10 UTC from IEEE Xplore. Restrictions apply.

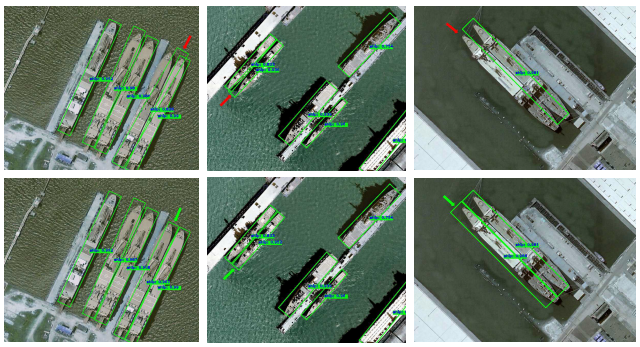


Fig. 18. (Top) Failure cases of the proposed method indicated by the red arrow. (Bottom) Modified detection results indicated by the green arrow.

as the final detection results by nonmaximum suppression, as shown in Fig. 17(c).

The Second Regional Proposal is not beneficial to detect the single ship on the sea surface. However, this method plays a positive effect on the detection of the closely arranged ships, since some missing targets are rediscovered and the bounding boxes of some targets are adjusted more perfect, as demonstrated in Fig. 18. The experimental results also confirm that the precision and recall rate on the HRSC2016 data set increase to 97.0% and 91.4%, respectively, and the average time improves to 0.384 s.

IV. Conclusion

Deep learning has achieved good performance in computer vision and image processing because it can automatically learn the essential features from large amounts of data. Based on the improvement of Faster R-CNN, a new framework for detecting the arbitrary direction ships in the remote-sensing image is proposed in this article, in which the bounding box of ship is determined by three sides, namely, vertical side, horizontal side, and short side, respectively. The inclination of ship is calculated by applying the arc-tangent operation between the vertical and horizontal sides. First, the better performing ResNet-101 network is used to extract features over the entire image, which will be shared by the RPN and the head network. Then, the multidirection proposal regions [Region of Interest (RoI)] that may contain the ship target are generated in the RPN, and RoIPooling is taken to unify the RoI feature size. For each generated RoI, the comprehensive features are obtained by combining the local and global features through the multiregion feature-fusion (MFF) module. Last, the head network uses the comprehensive features of the proposal regions for bounding box recognition by multitask learning, including classification, regression, and incline direction prediction (left or right).

Final experimental results validate that the proposed detection approach achieves competitive results and owns better robustness to the interference, such as dense arrangement, complicated background, and incomplete ships, compared with other ship-detection methods based on deep learning.

Acknowledgment

The authors would like to thank Dr. Z. Liu from the Chinese Academy of Sciences for his selfless delivery of the data set used in our experiment. The authors would also like to thank Prof. Z. Luo and Prof. F. Fang from the China University of Geosciences, Wuhan, for their helpful guidance and the anonymous scholars online for sharing the codes.

References

[1] F. Xu, J. Liu, M. Sun, D. Zeng, and X. Wang, "A hierarchical maritime target detection method for optical remote sensing imagery," *Remote Sens.*, vol. 9, no. 3, p. 280, Mar. 2017.

[2] T. Nie, B. He, G. Bi, Y. Zhang, and W. Wang, "A method of ship detection under complex background," *ISPRS Int. J. Geo-Inf.*, vol. 6, no. 6, p. 159, May 2017.

[3] G. Liu, Y. Zhang, X. Zheng, X. Sun, K. Fu, and H. Wang, "A new method on inshore ship detection in high-resolution satellite images using shape and context information," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 3, pp. 617–621, Mar. 2014.

[4] Z. Zou and Z. Shi, "Ship detection in spaceborne optical image with SVD networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 5832–5845, Jun. 2016.

[5] Q. Liu, X. Xiang, Y. Wang, Z. Luo, and F. Fang, "Aircraft detection in remote sensing image based on corner clustering and deep learning," *Eng. Appl. Artif. Intell.*, vol. 87, Jan. 2020, Art. no. 103333.

[6] H. He, Y. Lin, F. Chen, H.-M. Tai, and Z. Yin, "Inshore ship detection in remote sensing images via weighted pose voting," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3091–3107, Jun. 2017.

[7] S. Qi, J. Ma, J. Lin, Y. Li, and J. Tian, "Unsupervised ship detection based on saliency and S-HOG descriptor from optical satellite images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1451–1455, Jul. 2015.

[8] C. Zhu, H. Zhou, R. Wang, and J. Guo, "A novel hierarchical method of ship detection from spaceborne optical image based on shape and texture features," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 9, pp. 3446–3456, Sep. 2010.

[9] Y. Xia, S. Wan, and L. Yue, "A novel algorithm for ship detection based on dynamic fusion model of multi-feature and support vector machine," in *Proc. 6th Int. Conf. Image Graph.*, Aug. 2011, pp. 521–526.

[10] Y. D. Yu, X. B. Yang, S. J. Xiao, and J. L. Lin, "Automated ship detection from optical remote sensing images," *Key Eng. Mater.*, vol. 500, pp. 785–791, Jan. 2012.

[11] J. Antelo, G. Ambrosio, J. Gonzalez, and C. Galindo, "Ship detection and recognitionin high-resolution satellite images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, vol. 4, Jul. 2009, pp. 514–517.

[12] J. Xu, X. Sun, D. Zhang, and K. Fu, "Automatic detection of inshore ships in high-resolution remote sensing images using robust invariant generalized Hough transform," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 12, pp. 2070–2074, Dec. 2014.

[13] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 580–587.

[14] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[15] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[16] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[17] W. Liu *et al.*, "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.*, Oct. 2015, pp. 21–37.

[18] Y. Yao, Z. Jiang, H. Zhang, D. Zhao, and B. Cai, "Ship detection in optical remote sensing images based on deep convolutional neural networks," *J. Appl. Remote Sens.*, vol. 11, no. 4, p. 1, Sep. 2017.

[19] J. Li, C. Qu, and J. Shao, "Ship detection in SAR images based on an improved faster R-CNN," in *Proc. SAR Big Data Era, Models, Methods Appl. (BIGSARDATA)*, Nov. 2017, pp. 1–6.

[20] Y. Wang, C. Wang, H. Zhang, Y. Dong, and S. Wei, "A SAR dataset of ship detection for deep learning under complex backgrounds," *Remote Sens.*, vol. 11, no. 7, p. 765, Mar. 2019.

[21] Q. Li, M. Lichao, L. Qingjie, W. Yunhong, and Z. X. Xiang, "HSF-Net: Multiscale deep feature embedding for ship detection in optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 7147–7161, Jul. 2018.

[22] N. Bodla, B. Singh, R. Chellappa, and L. S. Davis, "Soft-NMS—Improving object detection with one line of code," 2017, *arXiv:1704.04503*. [Online]. Available: http://arxiv.org/abs/1704.04503

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:19:10 UTC from IEEE Xplore. Restrictions apply.

[23] R. Zhang, J. Yao, K. Zhang, C. Feng, and J. Zhang, "S-CNN-based ship detection from high-resolution remote sensing images," *ISPRS-Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 41, pp. 423–430, Jul. 2016.
[24] J. R. R. Uijlings, K. E. A. van de Sande, T. Gevers, and A. W. M. Smeulders, "Selective search for object recognition," *Int. J. Comput. Vis.*, vol. 104, no. 2, pp. 154–171, Sep. 2013.
[25] Z. Liu, J. Hu, L. Weng, and Y. Yang, "Rotated region based CNN for ship detection," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 900–904.
[26] Z. Liu, H. Wang, L. Weng, and Y. Yang, "Ship rotated bounding box space for ship extraction from high-resolution optical satellite images with complex backgrounds," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 8, pp. 1074–1078, Aug. 2016.
[27] H. Lin, Z. Shi, and Z. Zou, "Fully convolutional network with task partitioning for inshore ship detection in optical remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1665–1669, Oct. 2017.
[28] S. Zhang, R. Wu, K. Xu, J. Wang, and W. Sun, "R-CNN-Based ship detection from high resolution remote sensing imagery," *Remote Sens.*, vol. 11, no. 6, p. 631, Mar. 2019.
[29] S. Li, Z. Zhang, B. Li, and C. Li, "Multiscale rotated bounding box-based deep learning method for detecting ship targets in remote sensing images," *Sensors*, vol. 18, no. 8, p. 2702, Aug. 2018.
[30] X. Yang *et al.*, "Automatic ship detection in remote sensing images from Google Earth of complex scenes based on multiscale rotation dense feature pyramid networks," *Remote Sens.*, vol. 10, no. 1, p. 132, Jan. 2018.
[31] Y. Jiang *et al.*, "R2CNN: Rotational region CNN for orientation robust scene text detection," 2017, *arXiv:1706.09579*. [Online]. Available: http://arxiv.org/abs/1706.09579
[32] M. Liao and X. BaoguangandBai, "TextBoxes++: A single-shot oriented scene text detector," *IEEE Trans. Image Process.*, vol. 27, no. 8, pp. 3676–3690, Aug. 2018.
[33] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[34] J. Ma *et al.*, "Arbitrary-oriented scene text detection via rotation proposals," *IEEE Trans. Multimedia*, vol. 20, no. 11, pp. 3111–3122, Nov. 2018.
[35] W. Sun, K. Ren, G. Yang, X. Meng, and Y. Liu, "Investigating GF-5 hyperspectral and GF-1 multispectral data fusion methods for multitemporal change analysis," in *Proc. 10th Int. Workshop Anal. Multitemporal Remote Sens. Images (MultiTemp)*, Shanghai, China, Aug. 2019, pp. 1–4.
[36] M. Leordeanu, A. Radu, and R. Sukthankar, "Features in concert: Discriminative feature selection meets unsupervised clustering," Nov. 2014, *arXiv:1411.7714*. [Online]. Available: https://arxiv.org/abs/1411.7714
[37] S. Gidaris and N. Komodakis, "Object detection via a multi-region and semantic segmentation-aware CNN model," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1134–1142.
[38] M. Lin, Q. Chen, and S. Yan, "Network in network," 2013, *arXiv:1312.4400*. [Online]. Available: http://arxiv.org/abs/1312.4400
[39] Z. Liu, L. Yuan, L. Weng, and Y. Yang, "A high resolution optical satellite image dataset for ship recognition and some new baselines," in *Proc. 6th Int. Conf. Pattern Recognit. Appl. Methods*, 2017.
[40] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.
[41] G. Huang, "Computational investigation of wet area calculation of the large length-width ratio vessel," *Ship Eng.*, vol. 32, no. 3, pp. 17–19, Mar. 2010.
[42] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, Jun. 2010.
[43] D. M. W. E. Powers, "Evaluation: From predcision, recall and F-factor to ROC, informedness, markedness & correlation," *J. Mach. Learn. Technol.*, vol. 2, no. 1, pp. 37–63, 2011.
[44] S. Ruder, "An overview of multi-task learning in deep neural networks," 2017, *arXiv:1706.05098*. [Online]. Available: http://arxiv.org/abs/1706.05098

Qiangwei Liu received the B.S. degree in geographic information system from Ningxia University, Yinchuan, China, in 2015. He is pursuing the M.S. degree with the Intelligent System Software Technology Center, China University of Geosciences, Wuhan, China.

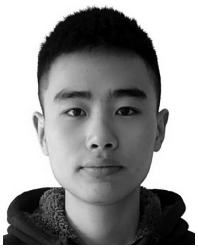
His research interests include deep learning and object detection.

Xiuqiao Xiang received the Ph.D. degree in systems analysis and integration from the Huazhong University of Science and Technology (HUST), Wuhan, China, in 2009.

From September 2009 to October 2011, he was a Mathematical Post-Doctoral Researcher with HUST. From August 2018 to December 2019, he was a Visiting Scholar with the Computer Vision Laboratory, University of California at Santa Barbara, Santa Barbara, CA, USA. He is an Associate Professor with the School of Geography and Information Engineering, China University of Geosciences, Wuhan. His research interests include remote sensing image processing, computer vision, machine learning, compressed sensing, fast transform, intelligent computation, and its application in information science.

Zhou Yang received the B.S. degree in remote sensing science and technology from Henan Polytechnic University, Jiaozuo, China, in 2016. He is pursuing the M.S. degree with the Intelligent System Software Technology Center, China University of Geosciences, Wuhan, China.

The main research interests of his postgraduate period are the processing and application of remote-sensing images.



Yu Hu received the B.S. degree in mechanical manufacturing and automation from Wuhan Donghu University, Wuhan, China, in 2018. He is pursuing the M.S. degree with Intelligent System Software Technology Center, China University of Geosciences, Wuhan.

His research interests include deep learning and image super-resolution reconstruction.

Yuming Hong received the B.S. degree in information engineering from the China University of Geosciences, Wuhan, China, in 2017, where he is pursuing the M.S. degree with the School of Geography and Information Engineering.

His research interests include deep learning and personalized tourist recommendation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 05:19:10 UTC from IEEE Xplore. Restrictions apply.